

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PNG 521 — Past Questions & Materials
by Renance

Course: PNG 521
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 500

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Paper 1 of 1 — PNG 521 ADVANCED DRILLING 2017 QUIZ

Level 500 • Second Semester • PastQuestion

2017 PNG Advance Drilling Engineering 50

Mat. No. _____ Mid-Semester Quiz Duration:

_____ 30 mins

1. Using the Fanning frictional pressure drop equation determine the bottomhole circulating pressure(BHCP) required to circulate drilling fluid 12 from the bottom of a 7800ft well being drilled with a 4" ID wrought iron drill pipe flowing 0.45 bbl/min of 1.25 specific gravity and 12 cp viscosity mud. The stand pipe pressure(PS) is 2150 psi while pressure drop across the bit is designed to be 400 psi for optimum bottomhole cleaning

a. $D = 4/12 = 0.333 \text{ ft}$

b. $U = Q = 0.45 \times 5.615 \times 4 = 10.167 \text{ ft/sec}$

$A = 60 \times 3.142 \times 0.03332$

c. $\rho = 1.25 \times 8.33 = 10.413 \text{ ppg}$

1

d. $N = 928 \text{ rpm}$ $d = 928 \times 10.413 \times 0.4835 \times 4 = 1,557.324$

e. Re

$\mu = 12$

2

f. $\square$ Flow is Laminar

1

g. Fanning friction factor $f = 16/NR = 16/1,557.324 = 0.0103$

0.0103

h. Frictional pressure drop using fanning: $L = 7800\text{ft}$, $Q = 0.45 \times 60 \times 24 = 648 \text{ bbl/day}$; $\rho = 1.25 \times 62.4 = 78 \text{ lb/ft}^3$; $D = 4"/12'/\text{ft} = 0.333\text{ft}$

0.333ft

L

$\Delta P_f = 1.8608 \times 10^{-9} f \cdot 5$

$\cdot \rho Q^2$

D

7800^3

$= 1.8608 \times 10^{-9} \times 0.0103 \cdot 5$

$\cdot 78 \times 648^2 = 1,195.8 \text{ psi}$

0.333

i. Total Frictional Pressure Drop $\square_{PT} = \square_{PPf} + \square_{PB} = 1,195.8 + 400 = 1,595.8\text{psi}$

j. $\square$ Annular pressure loss $= PS - \square_{PT} = 2150 - 1,595.8 = 554.21$

psi

$$k. \text{BHCP} = \text{PH} + \Delta P_{\text{ann}} = 0.052 \times 10.413 \times 7800 + 554.2 = 4,777.713 \text{ psi}$$

2. Which of the statements below adequately addresses flow regime

- The patterns in which the phases of materials move in conduit
- The different layers of fluid particles in moving through conduit
- The relationship between the Fluid Velocity & the Critical Velocity
- The character and behaviour of fluid during movement in conduit.

3. Determine the direction (in quadrant notation) of a well and its horizontal distance if its surface location is at N72000 and E61000 and the target location is at N21000 and E18000. N

$$\Delta N = 72,000 - 21,000 = 51,000$$

$$10 \Delta E = 61,000 - 18,000 = 43,000 \quad 10$$

Surface Location

7 W E

40.10

51,000 1

43,000

Target Location

S

6

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$\Delta E / W$

$$A = \tan^{-1} 2$$

$\Delta N / S$

43 , 000

$$= \tan^{-1} 0.843137$$

51 , 000 5

$$= 40.13550$$

Dire 2

ction

4. Define drilling service contract

in This is an agreement between the operator and the service

Qua 2

company to drill and complete a well OR to render certain

dran

specialized services within the drilling and completion processes.

t =

5.5 In drilling service contract the responsibility to supervise the operation to
40.guarantee successful delivery of the well belongs to

1 W a. The contractor that must ensure the well is drilled safely and
0

delivered 2

- b. The insurance provider for coverage of all accidents
- c. The owner of the well
- d. All of the above

6. Who CANNOT formulate the drilling service contract

- a. The operator
- b. The insurance provider and regulating bodies 2
- c. The service provider
- d. The association of the service providers

7. Mention four main types contracts used in the service industry of the oil
and gas industry

- a. Drilling contracts
- b. Well services contracts
- c. Seismic acquisition agreements

2

d. Master Service Contract

8. In payment clauses of drilling services contract, briefly explain the origin
of the "Footage Contract"

Stemed from the contract between Col. Drake and Mr. Smith to 2
drill the first oil well at Turtlesville and it was based on footage as
the depth where oil could be found was NOT known

9. Briefly explain the difference between Turnkey and Alliance Contracts

In the Turnkey contract, drilling contractor is obligated to drill a
complete well for a fixed fee and bears all risk But in Alliance, a 2
lead contractor is employed to subcontract and manage all the
services to drill a well, while the operator still supervises and
keep a close watch.

10.In liability or Risk Allocation Clauses of drilling service contract, what does
"Bury your own Dead" stand for and mention two areas/activities it does
NOT apply to

2

2

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Both operator and contractor separately assumes all risk

associated with its own equipment and personal, regardless of fault, and agrees to indemnify the other party for claims arising out of injuries suffered to their respective employees and properties.

11. What is the main difference between drilling method and drilling technology?

Drilling Methods consist of the body of techniques and equipment employed for rock breaking and sediment stirring into fragments or core in drilling (constructing or making hole) the well while Drilling Technology involves the application of other equipment, tools and drilling methods in the construction of a well

12. Briefly explain why rotary drilling is an abrasion drilling method

In abrasion drilling method, the rock is worn, gorged, peeled and chipped off by rotating action of a shoe or bit armed with some harder material buttons or teeth upon the rock. Rotary drilling is actually a technology that uses the Abrasion Drilling Method because it involves the rotation of the drillstring that has a bit at the bottom which crushes the rock which rotation and application of weight on the bit while circulating drilling fluid evacuate the drilled cuttings of the wellbore.

13. Figure below is the fracture and pore pressure gradient of formations to the reservoir at 14,000 ft.

- On the figure, give a preliminary casing design to drill to the reservoir.
- With casing made of steel of 65.4 b/ft³, calculate the Burst design pressure and Collapse design pressure using API Design factors

Surfa

16 ce

8

casin 4

g

1st 4800

Inter ft

2nd

medi

Inter

ate

medi Prod

casin

g uctio

ate

casin

9500
 g ft casin
 1150 g
 0 ft 14,0
 00 ft
 Prod 2
 uctio
 n
 casin
 g 2
 Burst
 desig 3
 n:
 Pore
 press